IBP for V++ Massive Bubble

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Abstract

In this note we construct IBP relations for the one-loop two-point (bubble) diagram in de Sitter space with both vertices being plus (V++) and internal propagators massive. We follow the structure of the reference note for V+- massive bubble, adding only the modifications required by the θ-function time-ordering. The momentum IBP relations remain unchanged, while time IBP relations acquire tadpole terms from derivatives of θ-functions. We provide the complete IBP system ready for Mathematica implementation.

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1 Background, goals, and structure

The goal is to construct practical reduction relations for the massive V++ de Sitter bubble, including the tadpole families induced by time ordering. Differential operators in external kinematics map integrals within the same families; after reduction to a master basis, these operators induce a first-order linear system for the master-integral vector.

We organize the note as follows. Section 2 fixes conventions and defines the $B^{(12)}$, $B^{(21)}$ and tadpole R families. Section 3 lists the IBP system and the kinematic differential operators. Section 4 records Baikov and dimension-shift relations for the momentum integral. Section 5 summarizes a $d\log$/intersection-theory viewpoint tailored to the present special-function setting.

2 Basic definitions, notation, and integral families

Notations The metric reads: $\mathrm{d}s^2 = a^2(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2)$ with scale factor $a(\tau) = -1/(H\tau)$ and $\tau \in (-\infty, 0)$. We set $H \equiv 1$ throughout. We use $\theta_{ij} \equiv \theta(\tau_i - \tau_j)$ and $\theta(0) = 1/2$. We work in D spatial dimensions (so the loop momentum integration is $\int \mathrm{d}^D\mathbf{q}$); in dimensional regularization one may take $D = 3-2\epsilon$. For simplicity we take all mass parameters equal, $\nu_1 = \nu_2 = \nu_3 = \nu_4 \equiv \nu$, and we also specialize to equal external energies $P_1 = P_2$.

Hankel building blocks Following the notation of arXiv:2401.00129v5 (Eqs. 3.11–3.12), we define the rescaled Hankel functions

$$h_\nu^{(1)}(x) \equiv x^{-\nu}\mathrm{H}_\nu^{(1)}(x), \qquad h_\nu^{(2)}(x) \equiv x^{-\nu}\mathrm{H}_{\nu^*}^{(2)}(x), \qquad \nu \in \mathbb{C}, \quad x > 0. \tag{1}$$

They satisfy (Eq. (3.13) in arXiv:2401.00129v5)

$$h''(x) + \frac{2\nu+1}{x}h'(x) + h(x) = 0. \tag{2}$$

The mass parameter is $m^2 = 9/4 - \nu^2$. We also define derivatives

$$h_{\nu,0}(x) \equiv h_\nu(x), \qquad h_{\nu,n}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n}h(x), \tag{3}$$

For the two branches we also write

$$h_{\nu,n}^{(1)}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n}h_\nu^{(1)}(x), \qquad h_{\nu,n}^{(2)}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n}h_\nu^{(2)}(x). \tag{4}$$

When the mass label is clear from context, we write $h_n \equiv h_{\nu,n}$ and similarly $h_n^{(1,2)} \equiv h_{\nu,n}^{(1,2)}$. In this note we consider a single mass scale, so we set all $\nu_i = \nu$ everywhere.

V++ bulk-to-bulk propagator and time ordering For a massive scalar on the + branch, the Schwinger–Keldysh propagator contains two time orderings,

$$B_\nu^{++}(k;\tau_1,\tau_2) \;\propto\; h_0^{(1)}(-k\tau_1)\,h_0^{(2)}(-k\tau_2)\,\theta_{12} \;+\; h_0^{(2)}(-k\tau_1)\,h_0^{(1)}(-k\tau_2)\,\theta_{21}, \tag{5}$$

with $\tau_1, \tau_2 \in (-\infty, 0)$. For the bubble, the product of two B^{++} propagators naturally splits into two contributions proportional to θ_{12} and θ_{21} respectively. This motivates treating the two orderings as two integral families related by symmetry.

Two integral families To clearly distinguish the two time-orderings, we define two integral families with explicit θ-function labels:

Family with θ_{12}

$$\begin{aligned} B^{(12)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \equiv \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2 \int \frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\,\frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ \times\, h_{n_1}^{(1)}(-q\tau_1)h_{n_2}^{(2)}(-q\tau_2) \\ \times\, h_{n_3}^{(1)}(-|\mathbf{q}+\mathbf{k}_s|\tau_1)h_{n_4}^{(2)}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\,\theta_{12}. \end{aligned} \tag{6}$$

Family with θ_{21}

$$\begin{aligned} B^{(21)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \equiv \int_{-\infty}^{0}\mathrm{d}\tau_1\mathrm{d}\tau_2 \int \frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\, e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2}\,\frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ \times\, h_{n_1}^{(2)}(-q\tau_1)h_{n_2}^{(1)}(-q\tau_2) \\ \times\, h_{n_3}^{(2)}(-|\mathbf{q}+\mathbf{k}_s|\tau_1)h_{n_4}^{(1)}(-|\mathbf{q}+\mathbf{k}_s|\tau_2)\,\theta_{21}. \end{aligned} \tag{7}$$

Relation between the two families The two families are related by exchanging time variables and external energies:

$$B^{(21)}[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\};P_1,P_2] = B^{(12)}[\{n_2,n_1,n_4,n_3\},\{a_2,a_1\},\{b_1,b_2\};P_2,P_1]. \tag{8}$$

Complete V++ correlator The physical V++ bubble correlator is the sum:

$$\mathcal{V}_{++} = B^{(12)} + B^{(21)}. \tag{9}$$

Tadpole families Time IBP relations involve total derivatives $\int \partial_{\tau_i}[\cdots] = 0$. When the integrand contains θ_{12} or θ_{21}, one generates coincidence terms proportional to $\delta(\tau_1-\tau_2)$, which reduce the two-time integral to a one-time integral. We treat these coincidence contributions as tadpole families R_1, R_2, which depend only on $P_0 \equiv P_1 + P_2$ (and we keep P_0 implicit in the notation).

$$\begin{aligned} R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] &\equiv e^{\pi \mathrm{Im}\,[\nu]}\frac{4i}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\,e^{-\mathrm{i}P_0\tau}\frac{(-\tau)^a}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ &\quad\times h^{(1)}_{n_3}(-|\mathbf{q}+\mathbf{k}_s|\tau)h^{(2)}_{n_4}(-|\mathbf{q}+\mathbf{k}_s|\tau), \end{aligned} \tag{10}$$

$$\begin{aligned} R_2[\{n_1,n_2\},\{a\},\{b_1,b_2\}] &\equiv e^{\pi \mathrm{Im}\,[\nu]}\frac{4i}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\,e^{-\mathrm{i}P_0\tau}\frac{(-\tau)^a}{|\mathbf{q}|^{b_1}|\mathbf{k}_s+\mathbf{q}|^{b_2}} \\ &\quad\times h^{(1)}_{n_1}(-q\tau)h^{(2)}_{n_2}(-q\tau). \end{aligned} \tag{11}$$

Direct Tadpole Family According to the Feynman rules, the tadpole diagram is given by the one-time integral of the equal-time propagator. Adopting the regularization $\theta(0) = 1/2$, the equal-time V++ propagator is

$$B^{++}_u(q;\tau,\tau) = \frac{1}{2}\left[h^{(1)}_u(-q\tau)h^{(2)}_u(-q\tau) + h^{(2)}_u(-q\tau)h^{(1)}_u(-q\tau)\right] = h^{(1)}_u(-q\tau)h^{(2)}_u(-q\tau). \tag{12}$$

We define the direct tadpole family T corresponding to this diagrammatic contribution:

$$\begin{aligned} T[\{n_1,n_2\},\{a\},\{b\}] &\equiv \int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\,e^{-\mathrm{i}P_0\tau}\frac{(-\tau)^a}{|\mathbf{q}|^{b}} \\ &\quad\times h^{(1)}_{n_1}(-q\tau)h^{(2)}_{n_2}(-q\tau). \end{aligned} \tag{13}$$

The families R_1, R_2 instead arise as remaining terms in time IBP for the bubble: one propagator is contracted using the Wronskian, producing the universal prefactor $e^{\pi \mathrm{Im}\,[\nu]}\frac{4i}{\pi}$ and an extra $(-k\tau)^{-2u-1}$ factor for the contracted line. In the integral-family language this is implemented as shifts of the τ-power and the corresponding $|k|$-power indices (e.g. $a \to a-2u-1$ and $b \to b+2u+1$), together with the constraint $\delta_{n_i+n_j,1}$ from the contraction formula.

3 Reduction relations for the one-loop two-point bubble

3.1 Top-sector families $B^{(12)}, B^{(21)}$

Origin of the tadpole (remaining) terms Time IBP relations involve total derivatives $\int \partial_{\tau_i}[\cdots] = 0$. With θ_{12} or θ_{21} present, derivatives generate coincidence terms proportional

to $\delta(\tau_1 - \tau_2)$. For Hankel-type integrands, the coincidence terms collapse one time integration using the Wronskian of the $h_\nu^{(1,2)}$ pair, producing one-time tadpole families. In the notation of arXiv:2401.00129v5 (Section 3.7), in the time derivation of B_{++}, the θ functions yield a remaining term

$$h_{n_1}^{(1)}(-k\tau_i)\, h_{n_2}^{(2)}(-k\tau_i) - h_{n_1}^{(2)}(-k\tau_i)\, h_{n_2}^{(1)}(-k\tau_i) = \delta_{n_1+n_2,1} e^{\pi \mathrm{Im}[\nu]} \frac{4i}{\pi} (-k\tau_i)^{-2\nu-1}\,. \tag{14}$$

which, in our integral-family language, is implemented as a shift of the corresponding $|k|$-power index by $2\nu+1$ and the τ-power index by $-2\nu-1$, together with the contraction of the two-time integrations by one integral in the remaining family.

In particular, since the R families originate from contracting one propagator in the top sector, the contracted line carries an extra $(-k)^{-2\nu-1}$ factor from the Wronskian. In the Mathematica/Kira implementation we keep the momentum baselines trivial,

$$|q|^{-b_1}\, |\boldsymbol{k}_s + \boldsymbol{q}|^{-b_2}, \qquad b_{10} = 0, \quad b_{20} = -2\nu,$$

and the remaining-term factor is tracked explicitly by the b shifts (e.g. $b_1 \to b_1 + 2\nu + 1$) in Eqs. (15)–(16).

Time IBP with θ-function tadpole terms In the following, B denotes $B^{(12)} + B^{(21)}$. The combination of the two parts can employ the propagator contraction formula, thereby simplifying the calculation. Momentum IBP and equation-of-motion relations apply to the families first. The only new ingredient in V++ (compare to V+-) is the θ function contributions in the time IBP.

For ∂_{τ_1}:

$$\begin{aligned} 0 = &- \mathrm{i}P_1 B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_1 B[\{n_1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1,b_2\}] \\ &- B[\{n_1+1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - B[\{n_1,n_2,n_3+1,n_4\},\{a_1,a_2\},\{b_1,b_2-1\}] \\ &+ \delta_{n_1+n_2,1}(-1)^{n_1+1} R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}] \\ &+ \delta_{n_3+n_4,1}(-1)^{n_3+1} R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}], \end{aligned} \tag{15}$$

and for ∂_{τ_2}:

$$\begin{aligned} 0 = &- \mathrm{i}P_2 B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_2 B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2-1\},\{b_1,b_2\}] \\ &- B[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - B[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2\},\{b_1,b_2-1\}] \\ &+ \delta_{n_1+n_2,1}(-1)^{n_2+1} R_1[\{n_3,n_4\},\{a_1+a_2-2\nu-1\},\{b_1+2\nu+1,b_2\}] \\ &+ \delta_{n_3+n_4,1}(-1)^{n_4+1} R_2[\{n_1,n_2\},\{a_1+a_2-2\nu-1\},\{b_1,b_2+2\nu+1\}], \end{aligned} \tag{16}$$

where ν is the common mass parameter.

The tadpole families R_1, R_2 are defined in Section 2.

Momentum IBP Momentum IBP relations are unchanged compared to the V+− bubble, because $\partial_{\boldsymbol{q}^i}\theta_{ij} = 0$. The derivation can be organized in the same chain-rule form as in Section 1.6 of the reference note by introducing

$$x_1 \equiv |\boldsymbol{q}|, \qquad x_2 \equiv |\boldsymbol{q} + \boldsymbol{k}_s|, \qquad \mathcal{O}_1 \equiv \boldsymbol{q}^i \partial_{\boldsymbol{q}^i}, \qquad \mathcal{O}_2 \equiv (\boldsymbol{q}^i + \boldsymbol{k}_s^i)\partial_{\boldsymbol{q}^i}, \tag{17}$$

whose actions on (x_1, x_2) are

$$\begin{aligned}
\mathcal{O}_1 x_1 &= x_1, \\
\mathcal{O}_1 x_2 &= \frac{1}{2} x_2^{-1}(x_1^2 + x_2^2 - k_s^2), \\
\mathcal{O}_2 x_1 &= \frac{1}{2} x_1^{-1}(x_1^2 + x_2^2 - k_s^2), \\
\mathcal{O}_2 x_2 &= x_2.
\end{aligned} \tag{18}$$

Equivalently, for any function $F(x_1, x_2)$ one has the chain rule

$$\mathcal{O}_\alpha F = (\mathcal{O}_\alpha x_1)\, \partial_{x_1} F + (\mathcal{O}_\alpha x_2)\, \partial_{x_2} F, \qquad \alpha = 1, 2. \tag{19}$$

From $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$ one may write an intermediate step in terms of the composite operator acting on the full integrand,

$$\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_1 \Big] \\
&\Bigg[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h_{n_1}^{(1)}(-x_1\tau_1) h_{n_2}^{(2)}(-x_1\tau_2) \times h_{n_3}^{(1)}(-x_2\tau_1) h_{n_4}^{(2)}(-x_2\tau_2) \Bigg] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_1 x_1)\partial_{x_1} + (\mathcal{O}_1 x_2)\partial_{x_2} \Big] \\
&\Bigg[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}}\, h_{n_1}^{(1)}(-x_1\tau_1) h_{n_2}^{(2)}(-x_1\tau_2) \times h_{n_3}^{(1)}(-x_2\tau_1) h_{n_4}^{(2)}(-x_2\tau_2) \Bigg].
\end{aligned} \tag{20}$$

Expanding the derivatives using $\partial_x x^{-b} = -(b/x)x^{-b}$ and $\partial_x h_n(-x\tau) = (-\tau)\, h_{n+1}(-x\tau)$ yields the closed relation in terms of shifted B integrals:

$$\begin{aligned}
0 = {} & D\, B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& - b_1\, B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_2 \Big(B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\
& \qquad + B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1 - 2, b_2 + 2\}] \\
& \qquad - k_s^2\, B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2 + 2\}] \Big) \\
& + B[\{n_1 + 1, n_2, n_3, n_4\}, \{a_1 + 1, a_2\}, \{b_1 - 1, b_2\}] \\
& + B[\{n_1, n_2 + 1, n_3, n_4\}, \{a_1, a_2 + 1\}, \{b_1 - 1, b_2\}] \\
& + \frac{1}{2} \Big(B[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2 - 1\}] \\
& \qquad + B[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1 + 2, b_2 + 1\}] \\
& \qquad - k_s^2\, B[\{n_1, n_2, n_3 + 1, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2 + 1\}] \Big) \\
& + \frac{1}{2} \Big(B[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1, b_2 - 1\}] \\
& \qquad + B[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2\, B[\{n_1, n_2, n_3, n_4 + 1\}, \{a_1, a_2 + 1\}, \{b_1, b_2 + 1\}] \Big),
\end{aligned} \tag{21}$$

where D is the spatial dimension (in dimensional regularization $D = 3 - 2\epsilon$). The second momentum IBP can be written in the same chain-rule form as

$$0 = \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \left[D + \mathcal{O}_2 \right] \left[e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{x_1^{b_1} x_2^{b_2}} h_{n_1}^{(1)}(-x_1\tau_1) h_{n_2}^{(2)}(-x_1\tau_2) \times h_{n_3}^{(1)}(-x_2\tau_1) h_{n_4}^{(2)}(-x_2\tau_2) \right], \tag{22}$$

and, after expansion, it gives the symmetric relation with the exchanges

$$\{n_1, n_2, n_3, n_4\} \leftrightarrow \{n_3, n_4, n_1, n_2\},$$
$$\{b_1, b_2\} \leftrightarrow \{b_2, b_1\}.$$

Equation of motion relations The Bessel equation (2) yields:

$$\begin{aligned} B[\{2, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- B[\{0, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\ &- (2\nu + 1) B[\{1, n_2, n_3, n_4\}, \{a_1 - 1, a_2\}, \{b_1 + 1, b_2\}], \end{aligned} \tag{23}$$

$$\begin{aligned} B[\{n_1, 2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- B[\{n_1, 0, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\ &- (2\nu + 1) B[\{n_1, 1, n_3, n_4\}, \{a_1, a_2 - 1\}, \{b_1 + 1, b_2\}], \end{aligned} \tag{24}$$

$$\begin{aligned} B[\{n_1, n_2, 2, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- B[\{n_1, n_2, 0, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \\ &- (2\nu + 1) B[\{n_1, n_2, 1, n_4\}, \{a_1 - 1, a_2\}, \{b_1, b_2 + 1\}], \end{aligned} \tag{25}$$

$$\begin{aligned} B[\{n_1, n_2, n_3, 2\}, \{a_1, a_2\}, \{b_1, b_2\}] = &- B[\{n_1, n_2, n_3, 0\}, \{a_1, a_2\}, \{b_1, b_2\}] \\ &- (2\nu + 1) B[\{n_1, n_2, n_3, 1\}, \{a_1, a_2 - 1\}, \{b_1, b_2 + 1\}], \end{aligned} \tag{26}$$

where ν is the common mass parameter.

Symmetry relations 1. **Propagator exchange**:

$$B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] = B[\{n_3, n_4, n_1, n_2\}, \{a_1, a_2\}, \{b_2, b_1\}]. \tag{27}$$

2. **Time exchange with energy swap**:

$$B^{(21)}[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = B^{(12)}[\{n_2, n_1, n_4, n_3\}, \{a_2, a_1\}, \{b_1, b_2\}; P_2, P_1]. \tag{28}$$

With the specialization $P_1 = P_2$, this becomes a direct symmetry relation between the two time-orderings at fixed kinematics:

$$B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_1] = B[\{n_2, n_1, n_4, n_3\}, \{a_2, a_1\}, \{b_1, b_2\}; P_1, P_1]. \tag{29}$$

Differential operators in kinematics The dependence on external energies is generated by derivatives with respect to P_1, P_2:

$$\frac{\partial}{\partial P_1} B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = i\, B[\{n_1, n_2, n_3, n_4\}, \{a_1 + 1, a_2\}, \{b_1, b_2\}; P_1, P_2], \tag{30}$$

$$\frac{\partial}{\partial P_2} B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}; P_1, P_2] = i\, B[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2 + 1\}, \{b_1, b_2\}; P_1, P_2]. \tag{31}$$

Derivative with respect to k_s We use

$$\frac{\partial}{\partial k_s}|\boldsymbol{q}+\boldsymbol{k}_s| = \frac{k_s^2+\boldsymbol{k}_s\cdot\boldsymbol{q}}{k_s|\boldsymbol{q}+\boldsymbol{k}_s|} = \frac{k_s^2+|\boldsymbol{q}+\boldsymbol{k}_s|^2-q^2}{2k_s|\boldsymbol{q}+\boldsymbol{k}_s|}, \tag{32}$$

which leads to a closed relation within the same B family:

$$\begin{aligned}
&\frac{\partial}{\partial k_s}B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}]\\
&= -\frac{b_2}{2k_s}\Big(k_s^2\,B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}]\\
&\qquad + B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}]\\
&\qquad - B[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}]\Big)\\
&+\frac{1}{2k_s}\Big(k_s^2\,B[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}]\\
&\qquad + B[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}]\\
&\qquad - B[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1-2,b_2+1\}]\Big)\\
&+\frac{1}{2k_s}\Big(k_s^2\,B[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}]\\
&\qquad + B[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}]\\
&\qquad - B[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}]\Big).
\end{aligned} \tag{33}$$

Scaling relation:

$$\begin{aligned}
&\left(k_s\frac{\partial}{\partial k_s}+P_1\frac{\partial}{\partial P_1}+P_2\frac{\partial}{\partial P_2}\right)B[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}]\\
&= (D-b_1-b_2-a_1-a_2-2)B[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}].
\end{aligned} \tag{34}$$

3.2 Tadpole families R_1, R_2

Tadpole families R_1, R_2 The tadpole terms R_1, R_2 are single-time integrals generated by the time IBP coincidence terms. They require their own IBP system for reduction.

Kira check: whether b_1 can be reduced to 0 todo

IBP relations for R_1 (and R_2 via symmetry) The tadpole family R_1 satisfies the following IBP relations. The corresponding relations for R_2 follow via the symmetry $\boldsymbol{q}\to-(\boldsymbol{q}+\boldsymbol{k}_s)$, which exchanges $b_1\leftrightarrow b_2$ and relabels Hankel indices appropriately.

Time IBP for R_1:

$$\begin{aligned}
0 =& -i(P_1+P_2)R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] - aR_1[\{n_3,n_4\},\{a-1\},\{b_1,b_2\}]\\
&- R_1[\{n_3+1,n_4\},\{a\},\{b_1,b_2-1\}] - R_1[\{n_3,n_4+1\},\{a\},\{b_1,b_2-1\}].
\end{aligned} \tag{35}$$

Momentum IBP for R_1: from $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots] = 0$,

$$
\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_1\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_1 x_1)\partial_{x_1} + (\mathcal{O}_1 x_2)\partial_{x_2}\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right]. \qquad (36)
\end{aligned}
$$

$$
\begin{aligned}
0 =& D R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] - b_1 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_2 \Big(R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] + R_1[\{n_3, n_4\}, \{a\}, \{b_1 - 2, b_2 + 2\}] \\
& - k_s^2 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2 + 2\}] \Big) \\
& + \frac{1}{2} \Big(R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2 R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 + 1\}] \Big) \\
& + \frac{1}{2} \Big(R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1 - 2, b_2 + 1\}] \\
& \qquad - k_s^2 R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 + 1\}] \Big). \qquad (37)
\end{aligned}
$$

Complementary momentum IBP for R_1: from $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i + \boldsymbol{k}_s^i) \cdots] = 0$,

$$
\begin{aligned}
0 &= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + \mathcal{O}_2\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right] \\
&= \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^D \boldsymbol{q}}{(2\pi)^D} \Big[D + (\mathcal{O}_2 x_1)\partial_{x_1} + (\mathcal{O}_2 x_2)\partial_{x_2}\Big] \left[e^{-\mathrm{i}P_0\tau} \frac{(-\tau)^a}{x_1^{b_1} x_2^{b_2}} h_{n_3}^{(1)}(-x_2\tau) h_{n_4}^{(2)}(-x_2\tau) \right]. \qquad (38)
\end{aligned}
$$

$$
\begin{aligned}
0 =& D R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] - b_2 R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - \frac{1}{2} b_1 \Big(R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] + R_1[\{n_3, n_4\}, \{a\}, \{b_1 + 2, b_2 - 2\}] \\
& - k_s^2 R_1[\{n_3, n_4\}, \{a\}, \{b_1 + 2, b_2\}] \Big) \\
& + R_1[\{n_3 + 1, n_4\}, \{a + 1\}, \{b_1, b_2 - 1\}] + R_1[\{n_3, n_4 + 1\}, \{a + 1\}, \{b_1, b_2 - 1\}]. \qquad (39)
\end{aligned}
$$

Equation of motion for R_1:

$$
\begin{aligned}
R_1[\{2, n_4\}, \{a\}, \{b_1, b_2\}] =& - R_1[\{0, n_4\}, \{a\}, \{b_1, b_2\}] \\
& - (2\nu + 1) R_1[\{1, n_4\}, \{a - 1\}, \{b_1, b_2 + 1\}], \qquad (40) \\
R_1[\{n_3, 2\}, \{a\}, \{b_1, b_2\}] =& - R_1[\{n_3, 0\}, \{a\}, \{b_1, b_2\}] \\
& - (2\nu + 1) R_1[\{n_3, 1\}, \{a - 1\}, \{b_1, b_2 + 1\}]. \qquad (41)
\end{aligned}
$$

Symmetry relations: R_1 inherits symmetry from the bubble:

$$
R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = R_1[\{n_4, n_3\}, \{a\}, \{b_1, b_2\}]. \qquad (42)
$$

More importantly, R_1 and R_2 are related by loop-momentum shift $\boldsymbol{q} \to -(\boldsymbol{q} + \boldsymbol{k}_s)$:

$$R_2[\{n_1, n_2\}, \{a\}, \{b_1, b_2\}] = R_1[\{n_1, n_2\}, \{a\}, \{b_2, b_1\}]. \tag{43}$$

This symmetry implies that one only needs to implement the IBP system for R_1; all relations for R_2 follow by applying $b_1 \leftrightarrow b_2$ and appropriate index relabeling.

These tadpole integrals are simpler than the double-time B integrals because they involve only one time variable τ and two Hankel functions (vs four in B), and can be reduced to master integrals using the above IBP system.

Differential operators for R_1 From the definition of R_1, we can derive relations by taking derivatives with respect to parameters:

Derivative with respect to P_0:

$$\frac{\partial}{\partial P_0} R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = i R_1[\{n_3, n_4\}, \{a+1\}, \{b_1, b_2\}]. \tag{44}$$

Derivative with respect to k_s:

$$\begin{aligned}
&\frac{\partial}{\partial k_s} R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
&= -\frac{b_2}{2k_s}\Big(k_s^2\, R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2+2\}] \\
&\qquad + R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] \\
&\qquad - R_1[\{n_3, n_4\}, \{a\}, \{b_1-2, b_2+2\}]\Big) \\
&+ \frac{1}{2k_s}\Big(k_s^2\, R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1, b_2+1\}] \\
&\qquad + R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1, b_2-1\}] \\
&\qquad - R_1[\{n_3+1, n_4\}, \{a+1\}, \{b_1-2, b_2+1\}]\Big) \\
&+ \frac{1}{2k_s}\Big(k_s^2\, R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1, b_2+1\}] \\
&\qquad + R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1, b_2-1\}] \\
&\qquad - R_1[\{n_3, n_4+1\}, \{a+1\}, \{b_1-2, b_2+1\}]\Big).
\end{aligned} \tag{45}$$

Scaling relation (dimensional analysis):

$$\left(k_s \frac{\partial}{\partial k_s} + P_0 \frac{\partial}{\partial P_0}\right) R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}] = (D - b_1 - b_2 - a - 1) R_1[\{n_3, n_4\}, \{a\}, \{b_1, b_2\}]. \tag{46}$$

This scaling relation will be used to verify the correctness of differential equations.

3.3 Odevity seclection rules for the IBP system

There are two useful parity classifications ("Odevity") for the bubble IBP system:

1. For dq-IBP and dτ-IBP, the parities of $n_1 + n_2 + b_1$ and $n_3 + n_4 + b_2$ are conserved.

2. For dq-IBP, the parities of $n_1 + n_3 + a_1$ and $n_2 + n_4 + a_2$ are conserved.

In reduction runs we restrict the IBP system to the even sector of the first classification, i.e.

$$n_1 + n_2 + b_1 \text{ even}, \qquad n_3 + n_4 + b_2 \text{ even}. \tag{47}$$

For the tadpole families this implies two independent constraints:

- For R_1 (obtained by contracting indices where one is 1 and the other is 0), one must impose b_1 even *and* $n_3+n_4+b_2$ even. This contraction produces a term with an odd power (excluding the cancelling 2ν shift), which matches the Odevity of the top sector.

- For R_2, the loop-momentum shift symmetry (43) exchanges $b_1 \leftrightarrow b_2$, so the corresponding restriction is b_2 even and $n_1 + n_2 + b_1$ even.

This is why Odevity belongs to the IBP chapter: it specifies which integrals are allowed to appear in the closed IBP system that we export to Kira.

3.4 Reduction strategy

1. Select a Odevity sub-system of IBP-system. Apply EoM and symmetry to time-IBP and momentum-IBP, Then feed the processed IBP relations to Kira for reduction, since Mathematica is too slow for the reduction step.

2. Reduce R_1, R_2 first. then, reduce B with the known R_1, R_2 in time IBP.

4 Baikov representation and dimension-shift relations

Baikov variables for the one-loop two-point momentum integral Introduce $x_1 \equiv |\boldsymbol{q}|$, $x_2 \equiv |\boldsymbol{q} + \boldsymbol{k}_s|$ and

$$z_1 = x_1^2 = q^2, \qquad z_2 = x_2^2 = (q + k_s)^2, \qquad \mathcal{K} = k_s^2, \tag{48}$$

with Gram determinant

$$\mathcal{G} = \frac{1}{4}\Big(2z_1k_s^2 + 2z_2k_s^2 - k_s^4 - z_1^2 - z_2^2 + 2z_1z_2\Big) = \frac{1}{4}\lambda(z_1, z_2, k_s^2), \tag{49}$$

where λ is the Källén function. The one-loop Baikov change of variables gives (for general d)

$$\mathrm{d}^d q \to C_d\, \mathcal{G}^{\frac{d-3}{2}} \mathcal{K}^{-\frac{d-2}{2}}\, \mathrm{d}z_1\, \mathrm{d}z_2, \qquad C_d = \frac{\pi^{-1/2}}{\Gamma[(d-1)/2]}. \tag{50}$$

In particular, since $\mathrm{d}z_i = 2x_i\, \mathrm{d}x_i$, one has the useful bookkeeping relation

$$B[\{\boldsymbol{n}\}, \{a_1, a_2\}, \{b_1 + 1, b_2 + 1\}] \sim \cdots \times \frac{\mathrm{d}z_1}{x_1^{b_0+b_1+1}} \frac{\mathrm{d}z_2}{x_2^{b_0+b_2+1}} = \cdots \times 4\, \frac{\mathrm{d}x_1}{x_1^{b_0+b_1}} \frac{\mathrm{d}x_2}{x_2^{b_0+b_2}}. \tag{51}$$

The integration region is determined by $\mathcal{G} \geq 0$, equivalently the triangle inequalities between (x_1, x_2, k_s).

Gram matrices and dimension-shift identity For a general L-loop integral with E independent external momenta, define the Gram matrices

$$(\mathcal{G}_{\text{full}})_{ij} \equiv q_i \cdot q_j, \qquad \{q_i\} = \{l_1, \ldots, l_L, p_1, \ldots, p_E\}, \tag{52}$$

where l_i are loop momenta and p_i are external momenta. The Gram determinant is $\mathcal{G} \equiv \det(\mathcal{G}_{\text{full}})$. The kinematic Gram matrix involves only external momenta:

$$\mathcal{K}_{ij} \equiv p_i \cdot p_j, \qquad i, j = 1, \ldots, E, \tag{53}$$

with $\mathcal{K} \equiv \det(\mathcal{K}_{ij})$.

Dimension shift With generalized Sylvester's determinant identity, one has

$$|\partial_{(q_i \cdot q_j)}|_{i,j \le L} |q_i \cdot q_j|^{\gamma} = |q_i \cdot q_j|_{i,j>L} \left(\prod_{i=0}^{L-1} (\gamma + i) \right) |q_i \cdot q_j|^{\gamma-1}. \tag{54}$$

For a symmetric matrix of scalar products one further has

$$\begin{aligned} &|A_{ij} \partial_{(q_i \cdot q_j)}|_{i,j \le L} |q_i \cdot q_j|^{\gamma} = |q_i \cdot q_j|_{i,j>L} \left(\prod_{i=0}^{L-1} \left(\gamma + \frac{i}{2} \right) \right) |q_i \cdot q_j|^{\gamma-1} \\ &A_{ij} = \frac{1}{2} \ \text{ for } \ i \neq j, \quad A_{ij} = 1 \ \text{ for } \ i = j. \end{aligned} \tag{55}$$

Here $|q_i \cdot q_j|_{i,j>L} = \mathcal{K}$ is the kinematic Gram determinant, and $|q_i \cdot q_j| = \mathcal{G}$ is the full Gram determinant. Writing

$$Q = \frac{1}{\prod_i z_i^{a_i}}, \qquad \gamma = \frac{d - E - L - 1}{2}, \tag{56}$$

the identity above leads to a dimension-raising relation in Baikov form:

$$\begin{aligned} C_{\gamma-1} \int Q \, \frac{\mathcal{G}^{\gamma-1}}{\mathcal{K}^{\gamma-1+L/2}} \prod \mathrm{d}z_i &= C_{\gamma} \int Q \, |A_{ij} \partial_{(q_i \cdot q_j)}|_{i,j \le L} \, \frac{\mathcal{G}^{\gamma}}{\mathcal{K}^{\gamma+L/2}} \prod \mathrm{d}z_i \\ &= (-1)^L C_{\gamma} \int Q \, |\overleftarrow{A_{ij} \partial_{(q_i \cdot q_j)}}|_{i,j \le L} \, \frac{\mathcal{G}^{\gamma}}{\mathcal{K}^{\gamma+L/2}} \prod \mathrm{d}z_i, \end{aligned} \tag{57}$$

where the left-hand side corresponds to a $d-2$ dimensional integral while the right-hand side produces a d dimensional one after integrating by parts.

5 Intersection theory viewpoint and dlog integrands

Convention In this section, we use the notation $\hat{F}$ to denote the integrand of a function F. For example, $\hat{B}$ is the integrand of the double-time integral B, and $\hat{I}$ is the integrand of the time integral I.

We define the shift parameters a_0 and b_0 based on the regularization and Hankel-function asymptotics. For the top-sector Bubble B, the time variables τ_1, τ_2 each carry a regularization

index δ_a. Additionally, we extract a factor $(-x_i\tau_i)^{-\nu}$ from each Hankel function to isolate the non-integer powers. With two Hankel functions per time variable, this adds 2ν to the τ weight and -2ν to the spatial weight. Thus we define:

$$\begin{aligned} a_0 &= \delta_a + 2\nu \quad (\text{for } B), \\ b_0 &= -2\nu. \end{aligned} \tag{58}$$

For the Tadpole family R_1, the structure arises from contracting one propagator in the Bubble (effectively setting $\tau_1 = \tau_2 = \tau$). The regularization indices sum up, $\delta_a + \delta_a = 2\delta_a$. The surviving propagator contributes the standard Hankel shift of 2ν. Thus for R_1 we use:

$$\begin{aligned} a_0 &= 2\delta_a + 2\nu \quad (\text{for } R_1), \\ b_0 &= -2\nu. \end{aligned} \tag{59}$$

In the following, the symbol a_0 refers to the shift parameter appropriate for the specific integral family under discussion.

By intersection theory, if a master-integrand basis ϕ_i is of $d\log$ form and the twist u satisfies $\mathrm{d}u = (\mathrm{d}\log u)\,\nu$, then the differential equations for the master integrals $\int \phi_i u$ take a $d\log$ form. Here we use a mild generalization motivated by fibration intersection theory: treat the master objects as $\Phi_i \cdot U$ with both Φ_i and U viewed as vectors, choose U to satisfy

$$\mathrm{d}U = \mathrm{d}\Omega \cdot U, \tag{60}$$

and construct Φ_i as $d\log$-form integrands. This motivates a canonical-like choice of masters for the de Sitter bubble.

Time-integral block as a first-order system Define the "tree-level-like" time integral (no momentum denominators)

$$I_{\boldsymbol{n}}(x_1, x_2; P_1, P_2) \equiv \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2\, \hat{I}_{\boldsymbol{n}}, \tag{61}$$

with the integrand

$$\begin{aligned} \hat{I}_{\boldsymbol{n}} \equiv & e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2}\,(-\tau_1)^{a_0}(-\tau_2)^{a_0} \\ & \times h^{(1)}_{n_1}(-x_1\tau_1) h^{(2)}_{n_2}(-x_1\tau_2)\, h^{(1)}_{n_3}(-x_2\tau_1) h^{(2)}_{n_4}(-x_2\tau_2). \end{aligned} \tag{62}$$

For fixed (ν, a_0), the $I_{\boldsymbol{n}}$ span a finite basis and satisfy a first-order system

$$\mathrm{d}I_{\boldsymbol{m}} = (\mathrm{d}\Omega_{\boldsymbol{mn}})\, I_{\boldsymbol{n}}, \tag{63}$$

where the one-forms in $\mathrm{d}\Omega$ are built from $\mathrm{d}\log x_1$, $\mathrm{d}\log x_2$ and $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$. For the two-vertex tree-level-like family entering the bubble, there are 2^4 time-integral master functions.

$$\mathrm{d}\Omega = \mathrm{d}\Omega_{x_1} + \mathrm{d}\Omega_{x_2} + \mathrm{d}\Omega_{ex}, \tag{64}$$

where $\mathrm{d}\Omega_{x_i}$ only contains $\mathrm{d}\log x_i$ while $\mathrm{d}\Omega_{ex}$ is composed of $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$ with constant coefficients (built from ν and a_0).

Baikov twist and dlog building blocks With ϵ the dimensional regulator ($d = 3 - 2\epsilon$), the one-loop Baikov measure contributes the factor $\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}$, and one can package the combined twist schematically as a vector

$$U = x_1^{-b_0} x_2^{-b_0}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\, I_{\boldsymbol{n}}, \tag{65}$$

whose differential is again a sum of $d\log$ one-forms,

$$\mathrm{d}U = \Big(\mathrm{d}\Omega - b_0(\mathrm{d}\log x_1 + \mathrm{d}\log x_2) - \epsilon(\mathrm{d}\log\mathcal{G} - \mathrm{d}\log\mathcal{K})\Big)\cdot U. \tag{66}$$

In terms of $z_i = x_i^2$, useful $d\log$ building blocks include

$$\begin{aligned} &\mathrm{d}\log z_1 = 2\mathrm{d}\log x_1, \qquad \mathrm{d}\log z_2 = 2\mathrm{d}\log x_2, \\ &\frac{\sqrt{z_2 k_s^2}\,\mathrm{d}z_1}{\mathcal{G}} = -\mathrm{d}\log\left(\frac{2\sqrt{z_2k_s^2} + k_s^2 - z_1 + z_2}{-2\sqrt{z_2k_s^2} + k_s^2 - z_1 + z_2}\right), \qquad z_1 \leftrightarrow z_2. \end{aligned} \tag{67}$$

Since $\mathrm{d}\int \mathrm{d}x_i/(P_i \pm x_1 \pm x_2)$ is of $d\log$ form, one also has the building blocks

$$\begin{aligned} &\partial_{P_j}\Omega_{ex}\,\mathrm{d}x_i = \partial_{P_j}\Omega\,\mathrm{d}x_i, \\ &\partial_{x_j}\Omega_{ex}\,\mathrm{d}x_i = (\partial_{x_j}\Omega - \partial_{x_j}\Omega_{x_j})\,\mathrm{d}x_i. \end{aligned} \tag{68}$$

Using

$$\begin{aligned} \hat{B}[\{\boldsymbol{n}\},\{a_1+1,a_2\},\{b_1,b_2\}] &= \partial_{P_1}\hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] = (\partial_{P_1}\Omega_{\boldsymbol{nm}}) \\ &\quad\times \hat{B}[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1,b_2\}], \\ \hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] &= \frac{1}{b_0+b_1-1}\,(\partial_{x_1}\Omega_{\boldsymbol{nm}}) \\ &\quad\times \hat{B}[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1-1,b_2\}], \end{aligned} \tag{69}$$

one may view $\partial_{P_j}\Omega\,\mathrm{d}x_i \sim \tau_j\,\mathrm{d}x_i$ and $\partial_{x_j}\Omega\,\mathrm{d}x_i \sim \frac{b_0+b_j-1}{x_j}\mathrm{d}x_j$ as additional $d\log$-type building blocks at the integrand level.

5.1 Top-sector $d\log$ candidates

Constructing candidate $d\log$ integrands For notational convenience, we split off common shifts and write

$$\hat{B}[\{\boldsymbol{n}\},\{a_0+a_1,a_0+a_2\},\{b_0+b_1,b_0+b_2\}] \to \hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}], \tag{70}$$

where $a_i, b_i \in \mathbb{Z}$ and (a_0, b_0) encode the universal shifts induced by the special-function normalization and prefactors.

Some candidate $d\log$-type integrands in $d = 3 - 2\epsilon$ are (up to overall constants)

$$\begin{aligned} &k_s\,\hat{B}[\{\boldsymbol{n}\},\{0,0\},\{2,2\};d=3] = \frac{\mathrm{d}x_1\,\mathrm{d}x_2}{x_1x_2}\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}, \\ &k_s\,\hat{B}[\{\boldsymbol{n}\},\{1,0\},\{2,1\};d=3] = \frac{\mathrm{d}x_1}{x_1}\,\tau_1\,\mathrm{d}x_2\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}, \\ &k_s\,\hat{B}[\{\boldsymbol{n}\},\{1,1\},\{1,1\};d=3] = \tau_1\,\mathrm{d}x_1\,\tau_2\,\mathrm{d}x_2\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \qquad \text{[CJQ: wrong, double pole]}. \end{aligned} \tag{71}$$

In $d = 1 - 2\epsilon$, one can also use the Baikov building block $\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}$ to form $d\log$ candidates such as

$$\begin{aligned}
\hat{B}[\{\boldsymbol{n}\},\{0,0\},\{1,0\};d=1] &= \frac{\mathrm{d}z_1}{z_1}\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
\hat{B}[\{\boldsymbol{n}\},\{1,0\},\{0,0\};d=1] &= \tau_1\,\mathrm{d}x_1\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I_{\boldsymbol{n}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned} \tag{72}$$

In addition, one may also consider candidates generated by Ω_{ex}:

$$\begin{aligned}
&(\partial_{x_2}\Omega_{ex})_{\boldsymbol{nm}}\,\mathrm{d}x_1\,\frac{\mathrm{d}x_2}{x_2}\,x_1^{-b_0}x_2^{-b_0}\,I_{\boldsymbol{m}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
&(\partial_{x_2}\Omega_{ex})_{\boldsymbol{nm}}\,\mathrm{d}x_1\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I_{\boldsymbol{m}}^{(R)}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned} \tag{73}$$

Master-integral selection for the bubble Based on the $d\log$ construction and IBP reduction, we select the following independent master integrals for the one-loop two-point bubble. The selection is performed at the original level (1-dimensional, without dimension raising), yielding 14 master integrals organized by dimension and index structure.

Master integrals at $d = 1$ (5 integrals):

$$\begin{aligned}
&B[\{0,0,0,1\},\{0,0\},\{0,1\},1], \quad B[\{1,1,0,1\},\{0,0\},\{0,1\},1],\\
&B[\{0,0,0,0\},\{0,1\},\{0,0\},1], \quad B[\{0,0,1,1\},\{0,1\},\{0,0\},1],\\
&B[\{1,1,1,1\},\{0,1\},\{0,0\},1].
\end{aligned} \tag{74}$$

Master integrals at $d = 3$ (9 integrals):

Type A –$\{0,1\},\{1,2\}$ index pattern:

$$\begin{aligned}
&B[\{0,1,0,0\},\{0,1\},\{1,2\},3], \quad B[\{0,1,1,1\},\{0,1\},\{1,2\},3],\\
&B[\{1,0,0,0\},\{0,1\},\{1,2\},3], \quad B[\{1,0,1,1\},\{0,1\},\{1,2\},3].
\end{aligned} \tag{75}$$

Type B –$\{0,0\},\{2,2\}$ $d\log$ form:

$$B[\{0,0,0,0\},\{0,0\},\{2,2\},3], \quad B[\{0,0,1,1\},\{0,0\},\{2,2\},3], \quad B[\{1,1,1,1\},\{0,0\},\{2,2\},3]. \tag{76}$$

These three integrals have the natural $d\log$ structure $k_s\,\hat{B}[\{n\},\{0,0\},\{2,2\};d=3] = \frac{\mathrm{d}x_1\,\mathrm{d}x_2}{x_1x_2}\cdots$.

Type C –linear combinations with prefactor $(2 + b_0 + 2\nu)$:

$$\begin{aligned}
&(2+b_0+2\nu)\,B[\{0,1,0,1\},\{0,0\},\{1,3\},3],\\
&(2+b_0+2\nu)\,B[\{0,1,1,0\},\{0,0\},\{1,3\},3].
\end{aligned} \tag{77}$$

The prefactor $(2 + b_0 + 2\nu)$ arises from the coefficient normalization during IBP reduction.

This master-integral basis is recorded in `codebubble/kira_1L2p/result/MIdlog.m` and serves as the starting point for constructing differential equations in external kinematics.

Raising $d=1$ back to $d=3$ For the Baikov measure one has

$$\mathrm{d}^d q \to C_d\, \mathcal{G}^{\frac{d-3}{2}} \mathcal{K}^{-\frac{d-2}{2}}\, \mathrm{d}z_1\, \mathrm{d}z_2. \tag{78}$$

In particular, for $d=1-2\epsilon$ and $d=3-2\epsilon$,

$$C_{1-2\epsilon}\, \mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} = \partial_{q^2}\left[C_{3-2\epsilon}\, \mathcal{K}^{-1/2+\epsilon}\mathcal{G}^{-\epsilon}\right], \tag{79}$$

where $\partial_{q^2} \equiv \partial_{z_1} + \partial_{z_2}$ and we used $\partial_{q^2}\mathcal{G} = \mathcal{K}$ together with $\Gamma(1-\epsilon) = -\epsilon\,\Gamma(-\epsilon)$.

Therefore, for

$$\begin{aligned}\hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= \mathrm{d}z_1\, \mathrm{d}z_2\, z_1^{-(b_0+b_1)/2} z_2^{-(b_0+b_2)/2}\, I_{\boldsymbol{n}}^{(R)} \\ &\quad \times (-\tau_1)^{a_1}(-\tau_2)^{a_2}\, C_{1-2\epsilon}\, \mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2},\end{aligned} \tag{80}$$

integration by parts gives

$$\begin{aligned}\hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= \mathrm{d}z_1\, \mathrm{d}z_2 \left[\left(\frac{b_0+b_1}{2z_1} + \frac{b_0+b_2}{2z_2}\right) I_{\boldsymbol{n}}^{(R)} - (\partial_{z_1}+\partial_{z_2})\, I_{\boldsymbol{n}}^{(R)}\right] \\ &\quad \times (-\tau_1)^{a_1}(-\tau_2)^{a_2}\, z_1^{-(b_0+b_1)/2} z_2^{-(b_0+b_2)/2}\, C_{3-2\epsilon}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.\end{aligned} \tag{81}$$

Equivalently, in (x_1,x_2) variables with $\partial_{q^2} = \frac{1}{2x_1}\partial_{x_1} + \frac{1}{2x_2}\partial_{x_2}$,

$$\begin{aligned}\hat{B}[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\};d=1] &= 2\,\mathrm{d}x_1\, \mathrm{d}x_2 \left[\left(\frac{b_0+b_1}{x_1^2} + \frac{b_0+b_2}{x_2^2}\right) I_{\boldsymbol{n}}^{(R)} - \left(\frac{1}{x_1}\partial_{x_1} + \frac{1}{x_2}\partial_{x_2}\right) I_{\boldsymbol{n}}^{(R)}\right] \\ &\quad \times (-\tau_1)^{a_1}(-\tau_2)^{a_2}\, x_1^{-(b_0+b_1-1)} x_2^{-(b_0+b_2-1)}\, C_{3-2\epsilon}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.\end{aligned} \tag{82}$$

In particular, for the $d=1$ candidates above,

$$\begin{aligned}\hat{B}[\{\boldsymbol{n}\},\{0,0\},\{1,0\};d=1] &= \mathrm{d}z_1\, \mathrm{d}z_2\, z_1^{-b_0/2-1/2} z_2^{-b_0/2}\, I_{\boldsymbol{n}}^{(R)}\, \mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} \\ &= 2\,\mathrm{d}x_1\, \mathrm{d}x_2 \left[\left(\frac{b_0+1}{x_1^2} + \frac{b_0}{x_2^2}\right) I_{\boldsymbol{n}}^{(R)} - \left(\frac{1}{x_1}\partial_{x_1} + \frac{1}{x_2}\partial_{x_2}\right) I_{\boldsymbol{n}}^{(R)}\right] \\ &\quad \times x_1^{-b_0} x_2^{-b_0+1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}, \\ \hat{B}[\{\boldsymbol{n}\},\{1,0\},\{0,0\};d=1] &= 2\,\mathrm{d}x_1\, \mathrm{d}x_2 \left[b_0\left(\frac{1}{x_1^2} + \frac{1}{x_2^2}\right) I_{\boldsymbol{n}}^{(R)} - \left(\frac{1}{x_1}\partial_{x_1} + \frac{1}{x_2}\partial_{x_2}\right) I_{\boldsymbol{n}}^{(R)}\right] \\ &\quad \times (-\tau_1)\, x_1^{-b_0+1} x_2^{-b_0+1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{-1/2+\epsilon}.\end{aligned} \tag{83}$$

Moreover,

$$\begin{aligned}&(\partial_{x_2}\Omega_{ex})_{\boldsymbol{nm}}\, \mathrm{d}x_1\, \frac{\mathrm{d}x_2}{x_2}\, x_1^{-b_0} x_2^{-b_0}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= (\partial_{x_2}\Omega - \partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\, \mathrm{d}x_1\, \frac{\mathrm{d}x_2}{x_2}\, x_1^{-b_0} x_2^{-b_0}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= \mathrm{d}x_1\, \mathrm{d}x_2\, (\partial_{x_2} I_{\boldsymbol{n}}^{(R)})\, x_1^{-b_0} x_2^{-b_0-1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} - \mathrm{d}x_1\, \mathrm{d}x_2\, (\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\, x_1^{-b_0} x_2^{-b_0-1}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= \mathrm{d}x_1\, (\mathrm{d}I_{\boldsymbol{n}}^{(R)})\, x_1^{-b_0} x_2^{-b_0-1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} - \mathrm{d}x_1\, \mathrm{d}x_2\, (\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\, x_1^{-b_0} x_2^{-b_0-1}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= -\mathrm{d}x_1\, I_{\boldsymbol{n}}^{(R)}\, \mathrm{d}\big(x_1^{-b_0} x_2^{-b_0-1}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\big) - \mathrm{d}x_1\, \mathrm{d}x_2\, (\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\, x_1^{-b_0} x_2^{-b_0-1}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= \left(\frac{b_0+1}{x_2}\delta_{\boldsymbol{nm}} - (\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\right) k_s\, x_1^{-b_0} x_2^{-b_0-1}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon-1/2}\, \mathrm{d}x_1\, \mathrm{d}x_2.\end{aligned}$$

and similarly

$$
\begin{aligned}
&(\partial_{x_2}\Omega_{ex})_{\boldsymbol{nm}}\,\mathrm{d}x_1\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I^{(R)}_{\boldsymbol{m}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\\
&=\left(\frac{b_0-1}{x_2}\delta_{\boldsymbol{nm}}-(\partial_{x_2}\Omega_{x_2})_{\boldsymbol{nm}}\right)x_1^{-b_0+1}x_2^{-b_0+1}\,I^{(R)}_{\boldsymbol{m}}\,\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2}\,\mathrm{d}x_1\,\mathrm{d}x_2.
\end{aligned}
$$

The practical goal is to pick a master basis for the full de Sitter bubble (including the special-function time dependence) such that the induced differential system in external kinematics is as close as possible to a $d\log$ form.

5.2 Alphabet and differential system structure

Based on the intersection-theory analysis, we have computed the differential equations for the master integrals. In the $d\log$ form, the differential system is characterized by an alphabet of letters. Unlike in flat-space calculations where the alphabet typically depends only on kinematic variables, in the massive de Sitter case with V++ vertices, the letters exhibit a non-trivial dependence on the mass parameter ν and the dimensional regulator ϵ.

The alphabet $\mathcal{A}$ appearing in the differential equations with respect to the kinematic variable x is found to be:

$$
\mathcal{A}=\left\{x,x\pm 1,x\pm\sqrt{1+2\epsilon},x\pm\frac{1+2\nu\pm\sqrt{3+4\epsilon+4\nu(1+\nu)}}{\sqrt{2}}\right\}. \tag{84}
$$

We observe that the alphabet contains square roots involving ϵ and ν, such as $\sqrt{1+2\epsilon}$ and $\sqrt{3+4\epsilon+4\nu(1+\nu)}$. This deviation from the rational structure usually found in flat-space polylogarithms is a hallmark of the massive de Sitter propagators and the specific time-ordering structure.

The coefficient matrix of the differential system, while too lengthy to be presented explicitly here, is constructed from rational functions of ϵ and ν, but decorated with the aforementioned square-root structures. These results for the alphabet and the structure of the differential equations provide the necessary data for expressing the master integrals in terms of iterated integrals, potentially generalizing the standard multiple polylogarithms to include algebraic kernels.

5.3 Tadpole-family $d\log$ candidates

Tadpole families R_1, R_2 and $d\log$ candidates The master integrals for the τ-dependent subfamily are denoted by $I^{(R)}_{\boldsymbol{n}}$, where the index vector $\boldsymbol{n}=\{n_3,n_4\}$ labels the Hankel function orders.

The tadpole terms R_1, R_2 can be treated similarly at the integrand level, by isolating the one-time "tree-level-like" block

$$
I^{(R)}_{\boldsymbol{n}}(x_2;P_0)\equiv\int_{-\infty}^{0}\mathrm{d}\tau\,\hat{I}^{(R)}_{\boldsymbol{n}}, \tag{85}
$$

where

$$
\hat{I}^{(R)}_{\boldsymbol{n}}\equiv e^{-\mathrm{i}P_0\tau}\,(-\tau)^{a_0}\,h^{(1)}_{n_3}(-x_2\tau)h^{(2)}_{n_4}(-x_2\tau). \tag{86}
$$

For fixed (ν, a_0), the $I_{\boldsymbol{n}}^{(R)}$ form a finite basis and satisfy a first-order system

$$\mathrm{d}I_{\boldsymbol{m}}^{(R)} = \left(\mathrm{d}\Omega_{\boldsymbol{mn}}^{(R)}\right) I_{\boldsymbol{n}}^{(R)}, \tag{87}$$

where $\mathrm{d}\Omega^{(R)}$ is built from $\mathrm{d}\log P_0$, $\mathrm{d}\log x_2$ and $\mathrm{d}\log(P_0 \pm 2x_2)$. The differential matrix decomposes as:

$$\mathrm{d}\Omega_{\boldsymbol{mn}}^{(R)} = (\Omega_{P_0})_{\boldsymbol{mn}} \,\mathrm{d}\log P_0 + (\Omega_{x_2})_{\boldsymbol{mn}} \,\mathrm{d}\log x_2 + (\Omega_-)_{\boldsymbol{mn}} \,\mathrm{d}\log\left(P_0 - 2x_2\right) + (\Omega_+)_{\boldsymbol{mn}} \,\mathrm{d}\log\left(P_0 + 2x_2\right), \tag{88}$$

with coefficient matrices:

$$\begin{aligned}
\Omega_{P_0} &= \frac{-a_0 + 2\nu}{2} - \frac{1+2\nu}{4}(\sigma_3 \otimes 1 + 1 \otimes \sigma_3) \\
&\quad + \frac{i(1+2\nu)}{4}(\sigma_1 \otimes \sigma_2 + \sigma_2 \otimes \sigma_1) + \frac{a_0 - 2\nu}{2}\sigma_2 \otimes \sigma_2, && (89)\\
\Omega_{x_2} &= -(1+2\nu) + \frac{1+2\nu}{2}(\sigma_3 \otimes 1 + 1 \otimes \sigma_3), && (90)\\
\Omega_+ &= \frac{-a_0 + 2\nu}{4}(1-\sigma_2) \otimes (1-\sigma_2) \\
&\quad + \frac{1+2\nu}{8}\left[(i\sigma_1 - \sigma_3) \otimes (1-\sigma_2) + (1-\sigma_2) \otimes (i\sigma_1 - \sigma_3)\right], && (91)\\
\Omega_- &= \frac{-a_0 + 2\nu}{4}(1+\sigma_2) \otimes (1+\sigma_2) \\
&\quad - \frac{1+2\nu}{8}\left[(i\sigma_1 + \sigma_3) \otimes (1+\sigma_2) + (1+\sigma_2) \otimes (i\sigma_1 + \sigma_3)\right]. && (92)
\end{aligned}$$

Furthermore, to study the asymptotic behavior at spatial infinity $x_2 \to \infty$, we can perform the variable transformation $x_2 = 1/t$. By doing so, we can extract the residue matrix Ω_∞ associated with the $t \to 0$ limit, which is defined as the coefficient of $\mathrm{d}\log t$. Substituting $x_2 = 1/t$ into the relevant differential forms yields:

$$\begin{aligned}
\mathrm{d}\log x_2 &= -\mathrm{d}\log t, \\
\mathrm{d}\log(P_0 \pm 2x_2) &= \mathrm{d}\log\left(\frac{P_0 t \pm 2}{t}\right) = \mathrm{d}\log\left(t \pm \frac{2}{P_0}\right) - \mathrm{d}\log t.
\end{aligned} \tag{93}$$

In the limit $t \to 0$, the $t \pm 2/P_0$ terms approach constants and do not contribute to the $\mathrm{d}\log t$ singularity. Thus, collecting the coefficients of $-\mathrm{d}\log t$ from $\mathrm{d}\Omega^{(R)}$, we obtain the definition of Ω_∞ as a simple linear combination:

$$\Omega_\infty = -\left(\Omega_{x_2} + \Omega_+ + \Omega_-\right). \tag{94}$$

This matrix Ω_∞ governs the simple pole at $x_2 \to \infty$.

The basis is formed by tensor products of Pauli matrices, where 1 denotes the 2×2 identity matrix. The matrix Ω_{x_2} is consistent with the integrability condition $d\Omega - \Omega \wedge \Omega = 0$. Note that $\partial_{P_0}\Omega$ contains a pole at $P_0 = 0$ from Ω_{P_0}. To construct a valid $d\log$ candidate for x_2 integration, one must subtract this contribution:

$$\left(\partial_{P_0}\Omega_{\boldsymbol{mn}}^{(R)} - \frac{1}{P_0}(\Omega_{P_0})_{\boldsymbol{mn}}\right)\mathrm{d}x_2 \sim \left(\tau\delta_{\boldsymbol{mn}} - \frac{\mathrm{i}}{P_0}(\Omega_{P_0})_{\boldsymbol{mn}}\right)\mathrm{d}x_2. \tag{95}$$

The combined twist can be packaged as

$$\nu_{\boldsymbol{n}}^{(R)} = x_1^{-b_0} x_2^{-b_0}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\, I_{\boldsymbol{n}}^{(R)}. \tag{96}$$

Some candidate $d\log$-type integrands for R_1 in $d = 3 - 2\epsilon$ are (up to overall constants)

$$\begin{aligned} k_s\, \hat{R}_1[\{\boldsymbol{n}\}, \{0\}, \{2,2\}; d=3] &= \frac{\mathrm{d}x_1\, \mathrm{d}x_2}{x_1 x_2}\, x_1^{-b_0} x_2^{-b_0}\, I_{\boldsymbol{n}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}, \\ -k_s\, \hat{R}_1[\{\boldsymbol{n}\}, \{1\}, \{2,1\}; d=3] - \mathrm{i}\frac{k_s}{P_0}(\Omega_{P_0})_{\boldsymbol{nm}} \hat{R}_1[\{\boldsymbol{m}\}, \{0\}, \{2,1\}; d=3] \\ = \frac{\mathrm{d}x_1}{x_1}\left(\tau\delta_{\boldsymbol{nm}} - \frac{\mathrm{i}}{P_0}(\Omega_{P_0})_{\boldsymbol{nm}}\right) \mathrm{d}x_2\, x_1^{-b_0} x_2^{-b_0}\, I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}. \end{aligned} \tag{97}$$

Besides the basic candidates above, one can still generate further $d\log$-type combinations from the $t = 1/x_2$ representation.

A further candidate from the x_2 equation.
Using $x_2 = 1/t$, the x_2 part of the differential system becomes

$$(\partial_{x_2} I_{\boldsymbol{m}}^{(R)})\, \mathrm{d}x_2 = \left(-\Omega_\infty \frac{\mathrm{d}x_2}{x_2} + \Omega_+ \mathrm{d}\log\left(t + \frac{2}{P_0}\right) + \Omega_- \mathrm{d}\log\left(t - \frac{2}{P_0}\right)\right)_{\boldsymbol{mn}} I_{\boldsymbol{n}}^{(R)}. \tag{98}$$

A further construction starting from $\hat{R}_1[\{\boldsymbol{n}\}, \{0\}, \{2,1\}; d=3]$.
Now start instead from

$$k_s\, \hat{R}_1[\{\boldsymbol{n}\}, \{0\}, \{2,1\}; d=3] = -\frac{\mathrm{d}x_1\, \mathrm{d}t}{x_1\, t^2}\, x_1^{-b_0} t^{b_0}\, I_{\boldsymbol{n}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}. \tag{99}$$

Performing IBP once more on the t variable gives

$$\begin{aligned} &(b_0 - 1)\, k_s\, \hat{R}_1[\{\boldsymbol{n}\}, \{0\}, \{2,1\}; d=3] \sim \frac{\mathrm{d}x_1\, \mathrm{d}t}{x_1\, t}\, x_1^{-b_0} t^{b_0}\, \partial_t I_{\boldsymbol{n}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\ &= \frac{\mathrm{d}x_1\, \mathrm{d}t}{x_1}\, x_1^{-b_0} t^{b_0} \left[(\Omega_\infty)_{\boldsymbol{nm}} \frac{1}{t^2} + (\Omega_+)_{\boldsymbol{nm}} \frac{1}{t\left(t + \frac{2}{P_0}\right)} + (\Omega_-)_{\boldsymbol{nm}} \frac{1}{t\left(t - \frac{2}{P_0}\right)}\right] I_{\boldsymbol{m}}^{(R)}\, \mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}. \end{aligned} \tag{100}$$

Here the first term is the non-$d\log$ obstruction, proportional to $\Omega_\infty\, \mathrm{d}t/t^2$. By contrast,

$$\begin{aligned} \frac{\mathrm{d}t}{t\left(t \pm \frac{2}{P_0}\right)} &= \pm\frac{P_0}{2}\left(\frac{\mathrm{d}t}{t} - \frac{\mathrm{d}t}{t \pm \frac{2}{P_0}}\right) \\ &= \pm\frac{P_0}{2}\left[\mathrm{d}\log t - \mathrm{d}\log\left(t \pm \frac{2}{P_0}\right)\right], \end{aligned} \tag{101}$$

or equivalently

$$\begin{aligned} \frac{2}{P_0}\frac{\mathrm{d}t}{t\left(t + \frac{2}{P_0}\right)} &= \mathrm{d}\log t - \mathrm{d}\log\left(t + \frac{2}{P_0}\right), \\ \frac{2}{P_0}\frac{\mathrm{d}t}{t\left(t - \frac{2}{P_0}\right)} &= -\mathrm{d}\log t + \mathrm{d}\log\left(t - \frac{2}{P_0}\right). \end{aligned} \tag{102}$$

Thus, after subtracting the $\Omega_\infty \,\mathrm{d}t/t^2$ piece and absorbing the overall normalization, one gets another valid $d\log$ construction:

$$
\begin{aligned}
&-\mathrm{i}\,\frac{2k_s}{P_0}\left((b_0-1)\delta_{\boldsymbol{nm}}-(\Omega_\infty)_{\boldsymbol{nm}}\right)\hat{R}_1[\{\boldsymbol{m}\},\{0\},\{2,1\};d=3]\\
&=-\mathrm{i}\,\frac{\mathrm{d}x_1\,\mathrm{d}t}{x_1}\,x_1^{-b_0}t^{b_0}\left[(\Omega_+)_{\boldsymbol{nm}}\frac{2/P_0}{t\left(t+\frac{2}{P_0}\right)}+(\Omega_-)_{\boldsymbol{nm}}\frac{2/P_0}{t\left(t-\frac{2}{P_0}\right)}\right]I^{(R)}_{\boldsymbol{m}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned}
\tag{103}
$$

The two kernels on the right-hand side are now directly normalized as linear combinations of $\mathrm{d}\log t$ and $\mathrm{d}\log(t\pm 2/P_0)$, with coefficients independent of the external kinematics.

In $d=1-2\epsilon$, one can also use the Baikov building blocks $\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}$ and $\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}$ to form additional $d\log$ candidates such as

$$
\begin{aligned}
\hat{R}_1[\{\boldsymbol{n}\},\{0\},\{1,0\};d=1]&=\frac{\mathrm{d}z_1}{z_1}\,\frac{k_s\sqrt{z_1}\,\mathrm{d}z_2}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I^{(R)}_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
\hat{R}_1[\{\boldsymbol{n}\},\{0\},\{0,1\};d=1]&=\frac{\mathrm{d}z_2}{z_2}\,\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I^{(R)}_{\boldsymbol{n}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon},\\
-\hat{R}_1[\{\boldsymbol{n}\},\{1\},\{0,0\};d=1]&-\frac{\mathrm{i}}{P_0}(\Omega_{P_0})_{\boldsymbol{nm}}\hat{R}_1[\{\boldsymbol{n}\},\{0\},\{0,0\};d=1]\\
&=\left(\tau\delta_{\boldsymbol{nm}}-\frac{\mathrm{i}}{P_0}(\Omega_{P_0})_{\boldsymbol{nm}}\right)\mathrm{d}x_2\,\frac{k_s\sqrt{z_2}\,\mathrm{d}z_1}{\mathcal{G}}\,z_1^{-b_0/2}z_2^{-b_0/2}\,I^{(R)}_{\boldsymbol{m}}\,\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}.
\end{aligned}
\tag{104}
$$

6 Differential equations for the tadpole master integrals

6.1 Master-integral definitions

The Kira implementation uses shifted indices

$$
(-\tau)^{a_0+a},\qquad |q|^{-b_1},\qquad |\boldsymbol{q}+\boldsymbol{k}_s|^{-b_2},
\tag{105}
$$

with shifts (consistent with the conventions in Section 5)

$$
b_{10}=0,\qquad b_{20}=-2\nu,\qquad D=3-2\epsilon.
\tag{106}
$$

Note on $d\log$ indices: The R-function representation uses x-variables ($z=x^2$). The $d\log$ condition $dz/z=2dx/x$ implies an integrand behavior of $1/x^2$ (index 2) in the R-representation, compared to $1/z$ (index 1) in the Baikov representation. Thus, $d\log$ basis elements in this setup correspond to indices $b_i=2$, not 1.

In this convention the tadpole family is

$$
\begin{aligned}
R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}]\equiv{}& e^{\pi\mathrm{Im}\,[\nu]}\frac{4i}{\pi}\int_{-\infty}^{0}\mathrm{d}\tau\int\frac{\mathrm{d}^D\mathbf{q}}{(2\pi)^D}\,e^{-\mathrm{i}P_0\tau}\\
&\times\frac{(-\tau)^{a_0+a}}{|q|^{b_{10}+b_1}\,|\boldsymbol{q}+\boldsymbol{k}_s|^{b_{20}+b_2}}h^{(1)}_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau)h^{(2)}_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau).
\end{aligned}
\tag{107}
$$